Learning correspondence estimation for deformable 3D shapes

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by

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Spring Semester, 2025-26

8 Nov 2026

DECLARATION

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CERTIFICATE

This is to certify that the project report entitled "**Learning correspondence estimation for deformable 3D shapes**" submitted by **Lubesh Kumar Sharma** (Roll No. 22CS30065) to Indian Institute of Technology Kharagpur towards partial fulfilment of requirements for the award of degree of Integrated Dual Degree (B.Tech + M.Tech) in Computer Science and Engineering is a record of bona fide work carried out by him under my supervision and guidance during Spring Semester, 2025-26.

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Abstract

Name of the student: **Lubesh Kumar Sharma** Roll No: **22CS30065**

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Thesis title: **Learning correspondence estimation for deformable 3D shapes**

Thesis supervisor: **Professor Ayan Chaudhury**

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Dense correspondence between deformable three-dimensional shapes requires a dense map that preserves anatomical meaning across changes in pose and orientation. This report presents a point-cloud pipeline combining spectral geometry, relative orientation estimation, masked representation learning and deformation-guided matching. Heat-based descriptors and explicit coordinates describe each point, while an orientation module adapted from SE-ORNet predicts relative yaw through overlapping angular bins and continuous refinement. A shared Adaptive Siamese Masked Autoencoder learns geometric features through cross-shape reconstruction, supported by intrinsic consistency, self-identity supervision and confidence-weighted DINOv2 distillation.

Heat-Anchored Deformation Matching combines the learned geometric and visual features with intrinsic anchor verification, anchor-relative descriptors and non-rigid

deformation. The resulting dense map supports attribute transfer between differently posed shapes. The experimental progression examines coordinate conditioning, visual cues, distortion regularization, point-estimated normals, canonicalization and relative orientation. Qualitative observations of partial left–right improvement under distortion regularization motivated further study of front–back alignment. The report describes the roles of the pipeline modules and the experimental rationale connecting them, with a concise overview following the conclusion and a detailed account in the appendix.

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Abbreviations

3D	**T**hree-**D**imensional
AAMG	**A**daptive **A**dversarial **M**ask **G**enerator
AMIGO	**A**daptive **Siamese** **M**asked **A**utoencoder with GO
ARAP	**A**s-**R**igid-**A**s-**P**ossible
ASMAE	**A**daptive **S**iamese **M**asked **A**utoencoder
CD	**C**hamfer **D**istance
CE	**C**ross-**E**ntropy
CoE	**C**oupled **E**mbedding
CVPR	**C**onference on **C**omputer **V**ision and **P**attern **R**ecognition
DGCNN	**D**ynamic **G**raph **C**onvolutional **N**eural **N**etwork
DPC	**D**eep **P**oint **C**orrespondence
DV-Matcher	**D**eformation-based **N**on-**r**igid **P**oint **C**loud **M**atching
DINOv2	Self-supervised visual feature backbone
EMA	**E**xponential **M**oving **A**verage
FIM	**F**eature **I**nteraction **M**odule
FPS	**F**arthest **P**oint **S**ampling
GELU	**G**aussian **E**rror **L**inear **U**nit
GO	**G**lobal **O**ptimization
HADM	**H**eat-**A**nchored **D**eformation **M**atching
HKS	**H**eat **K**ernel **S**ignature
InfoNCE	Noise-contrastive representation-learning objective
ICCV	**I**nternational **C**onference on **C**omputer **V**ision
kNN	**k**-**N**earest **N**eighbor
LBO	**L**aplace-**B**eltrami **O**perator
LG-NET	**L**ocal-and-**G**lobal attention **N**etwork
LiDAR	**L**ight **D**etection **A**nd **R**anging
MAE	**M**asked **A**utoencoder
MGE	**M**ean **G**eodesic **E**rror
MLP	**M**ulti-**L**ayer **P**erceptron

NeRF	**N**eural **R**adiance **F**ields
OEM	**O**rientation **E**stimation **M**odule
OT	**O**ptimal **T**ransport
PCA	**P**rincipal **C**omponent **A**nalysis
p2p	**P**oint-**to**-**P**oint
SCAPE	**S**hape **C**ompletion **a**nd **A**nimation for **P**eople
SE-ORNet	**S**elf-**E**nsembling **O**rientation-**A**ware **N**etwork
WKS	**W**ave **K**ernel **S**ignature
SHOT	**S**ignature of **H**istograms of **O**rientations

Chapter 1

Introduction

1.1 Problem and Scope

Dense correspondence establishes a point-to-point relationship between two three-dimensional shapes. For deformable human shapes, the same anatomical region can occupy very different positions as the body changes pose. A useful correspondence must therefore combine local shape information with the spatial relationships between body parts. This is important for transferring surface attributes, comparing shapes and analysing deformation.

Point clouds provide a direct representation of a surface through its sampled positions. Estimating correspondence from these positions is challenging because local regions can have similar geometry, while global orientation and articulation change their coordinates. Bilateral symmetry introduces left–right ambiguity, and differences in facing direction introduce front–back ambiguity. These observations motivate a system that combines geometric descriptors, orientation estimation, learned point features and deformation-guided matching.

The proposed pipeline uses an Adaptive Siamese Masked Autoencoder (ASMAE) to learn point representations from heat-based descriptors and coordinates. A relative

orientation module aligns the input pair, while DINOv2 supplies complementary visual information. Heat-Anchored Deformation Matching (HADM) integrates these features with intrinsic geometric consistency and non-rigid refinement to obtain the final dense correspondence.

Chapter 2

Related Works

2.1 Learning Correspondence from Point Clouds

PointNet (Qi et al., 2017a) learns from unordered point sets through shared pointwise operations and symmetric aggregation. PointNet++ (Qi et al., 2017b) introduces hierarchical neighbourhood processing, while DGCNN (Wang et al., 2019) constructs edge features on learned neighbourhoods. These methods provide a foundation for extracting local and global information directly from point coordinates.

DPC (Lang et al., 2021) learns correspondence through cross- and self-construction of point sets. CorrNet3D (Zeng et al., 2021) also studies unsupervised dense point-cloud correspondence. Reconstruction provides a training signal without annotated inter-shape matches, connecting the learned representation to the geometry of the input shapes. Our representation-learning module follows this general principle and uses paired masked reconstruction.

2.2 Masked Reconstruction and ASMAE

Point-MAE (Pang et al., 2022) learns point-cloud representations by reconstructing masked patches. Point-M2AE (Zhang et al., 2022) extends masked learning to multiple spatial scales. AMIGO (Deng and Lu, 2025) introduces an Adaptive Siamese Masked Autoencoder that reconstructs source information using a target reference, together with adaptive masking and a global optimal-transport objective.

Our ASMAE uses shared encoding and cross-attention reconstruction to connect the two shapes. Stochastic masking hides selected descriptor entries while preserving their coordinate context. Clean point embeddings also support geometric and visual supervision. The teacher-based variants described in the appendix draw on exponential-moving-average consistency learning (Tarvainen and Valpola, 2017), and their transport formulation is related to entropically regularized matching (Cuturi, 2013).

2.3 Spectral Geometry and Intrinsic Distances

The Heat Kernel Signature (HKS) (Sun et al., 2009) characterizes shape points through heat diffusion. The Wave Kernel Signature (WKS) (Aubry et al., 2011) uses spectral energy bands to provide a complementary signature. Both relate point descriptors to the spectrum of a geometric operator.

Our heat representation samples responses between each point and a set of landmarks. These responses form a point-by-landmark descriptor matrix. Classical HKS samples the diagonal of the heat kernel across diffusion times; landmark responses instead describe diffusion relative to selected source locations. The common heat-kernel foundation supplies spatially structured spectral information for point representation learning.

Biharmonic distance (Lipman et al., 2010) provides a spectral measure of separation between shape points. Our method uses a biharmonic embedding for metric regularization and anchor verification. CoE (Zeng et al., 2025) provides related motivation for learning coupled embeddings with geometric structure. Intrinsic distance information complements coordinate-based alignment by describing relationships across the shape.

2.4 Relative Orientation Estimation

Shape–Pose Disentanglement (Katzir et al., 2022) studies the separation of shape and pose using equivariant representations. Continuous rotation parameterizations (Zhou et al., 2019) provide a way to predict valid rotations without discontinuities introduced by some angular representations. These ideas motivate the canonical-orientation variants in our experimental progression.

SE-ORNet (Deng et al., 2023) combines orientation estimation, domain adaptation and self-ensembling for unsupervised correspondence. Its pairwise orientation module motivates our use of synthetic rotations to supervise relative alignment. Gradient reversal (Ganin et al., 2016) transfers the learned representation between synthetic same-shape pairs and pairs of different shapes. Group normalization (Wu and He, 2018) normalizes intermediate features within the orientation network.

MultiBin (Mousavian et al., 2017) combines coarse angular classification with continuous residual prediction. We adapt this principle to relative yaw between point clouds. Overlapping bins represent multiple nearby angular hypotheses, and a residual provides the displacement from the chosen bin centre. Consistency across independently sampled point subsets supports stable predictions from different observations of the same pair.

2.5 Visual Features and Deformation Matching

DINOv2 (Oquab et al., 2024) provides pretrained visual representations that can be transferred from rendered views to three-dimensional points. I2P-MAE (Zhang et al., 2023) also explores image-to-point representation learning. DV-Matcher (Chen et al., 2025) motivates combining visual features with deformation constraints for non-rigid matching. Our visual branch lifts image descriptors to visible points and uses them both for feature distillation and correspondence estimation.

The self-identity objective follows the InfoNCE formulation of contrastive predictive coding (van den Oord et al., 2018). Corresponding point identities in two augmented observations of the same shape provide positive pairs. This trains representations to preserve point identity under changes in observation and geometry.

Spectral matching (Leordeanu and Hebert, 2005) evaluates candidate correspondences through pairwise compatibility and a leading eigenvector. HADM uses this principle to score anchor candidates by agreement of intrinsic distances. Anchor-relative descriptors then propagate geometric context to the remaining points.

Embedded deformation (Sumner et al., 2007) represents non-rigid motion through transformations attached to a graph. As-rigid-as-possible regularization (Sorkine and Alexa, 2007) encourages compatible local transformations. HADM optimizes a deformation graph during matching, using point features and anchors to guide the fit. The combination of visual features, intrinsic anchor consistency and deformation supplies complementary evidence for dense assignment.

Chapter 3

Methodology

3.1 Pipeline Overview

Let $X = \{x_i\}_{i=1}^{N}$ and $Y = \{y_j\}_{j=1}^{M}$ be source and target point clouds. Dense correspondence assigns each source point a target index through a map $T : \{1, \ldots, N\} \to \{1, \ldots, M\}$. Figure 3.1 presents the modules that compute this map and the supervision used to learn their representations.

Preprocessing converts each shape into a point cloud with heat-based descriptors and coordinate channels. The orientation module estimates relative yaw and aligns the target. A shared ASMAE learns geometric point features through masked cross-shape reconstruction. The visual branch extracts DINOv2 descriptors from projected views and transfers their information to the learned point representation. HADM fuses the geometric and visual features, identifies consistent anchors and refines the source through non-rigid deformation before dense assignment.

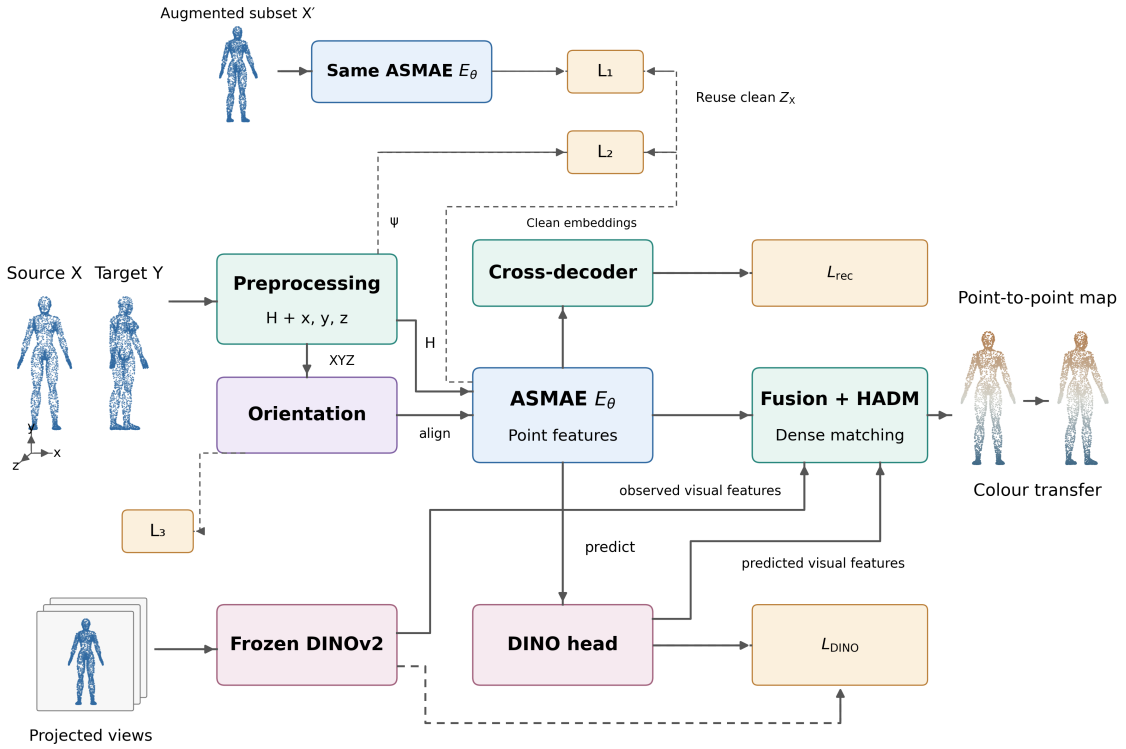


FIGURE 3.1: Correspondence pipeline and training objectives. Dashed connections identify supervision; solid connections show the flow of features and geometry. Here $L_1 = L_{\text{self}}$ is the self-identity objective; L_2 groups L_{cycle}, L_B, L_{orient} and L_{sep}; and L_3 groups the orientation objectives L_{bin}, L_{res}, L_{subset} and L_{domain}. The colour-transfer pair illustrates how a dense point map transfers a source attribute to its corresponding target points.

3.2 Point-Cloud Preprocessing

3.2.1 Surface points and neighbourhood graph

The preprocessing module extracts vertex positions from the input shape and represents them as an unordered point cloud. Coordinates are centred and normalized to provide a consistent scale. A nearest-neighbour graph connects spatially nearby points, and its undirected adjacency matrix A defines the graph Laplacian

$$L = D - A, \tag{3.1}$$

where D is the diagonal degree matrix. Eigenpairs $(\lambda_\ell, \phi_\ell)$ describe variation over the graph and provide the spectral basis for the geometric descriptors. Figure 3.2 shows the conversion from surface geometry to the input feature matrix.

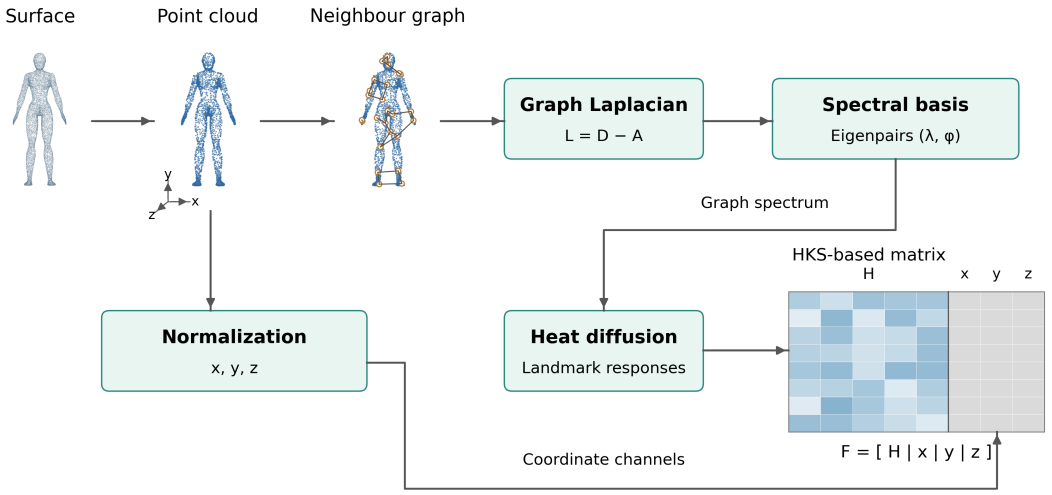


FIGURE 3.2: Point-cloud and heat-descriptor preprocessing. Sampled surface positions form the point cloud and neighbourhood graph. The graph spectrum produces the heat-based descriptor matrix H, which is concatenated with normalized x, y and z coordinates. Matrix shading illustrates the organization of descriptor channels.

3.2.2 Heat descriptors and coordinate channels

Heat diffusion describes how information spreads between locations on the graph. For sampled landmarks $s_1, \ldots, s_q$, the descriptor at point x_i is

$$H_{ij} = K_t(x_i, s_j) = \sum_\ell e^{-t\lambda_\ell} \phi_\ell(x_i)\phi_\ell(s_j). \tag{3.2}$$

This HKS-based representation uses landmark heat responses (Sun et al., 2009). Each row corresponds to a point, and each column records diffusion from a landmark. Channel normalization and logarithmic scaling balance the descriptor magnitudes.

The encoder input concatenates the heat matrix with explicit coordinates:

$$F_X = [H_X \mid x \mid y \mid z]. \tag{3.3}$$

The coordinate channels supply position information alongside the spectral descriptor. They are expressed in the same frame as the point positions passed to the encoder. After relative alignment, the target coordinate channels are updated accordingly.

3.2.3 Intrinsic geometric embedding

A biharmonic embedding supplies distances for geometric regularization and anchor consistency (Lipman et al., 2010):

$$\psi(x) = \left[\frac{\phi_1(x)}{|\lambda_1|}, \ldots, \frac{\phi_r(x)}{|\lambda_r|}\right], \qquad d_B(x, x') = \|\psi(x) - \psi(x')\|_2. \tag{3.4}$$

Nonzero spectral modes define the embedding, which is normalized by a reference distance scale. The resulting representation describes the relative separation of points through the graph geometry.

3.3 Relative Orientation Module

3.3.1 Pairwise encoding and feature interaction

The orientation estimation module (OEM) takes independently sampled subsets of the source and target. Shared point encoding extracts local features, and cross-shape feature interaction relates regions of one shape to the other. Global pooling produces a pair representation for predicting relative yaw about the vertical axis.

The architecture follows the orientation-learning principle of SE-ORNet (Deng et al., 2023).

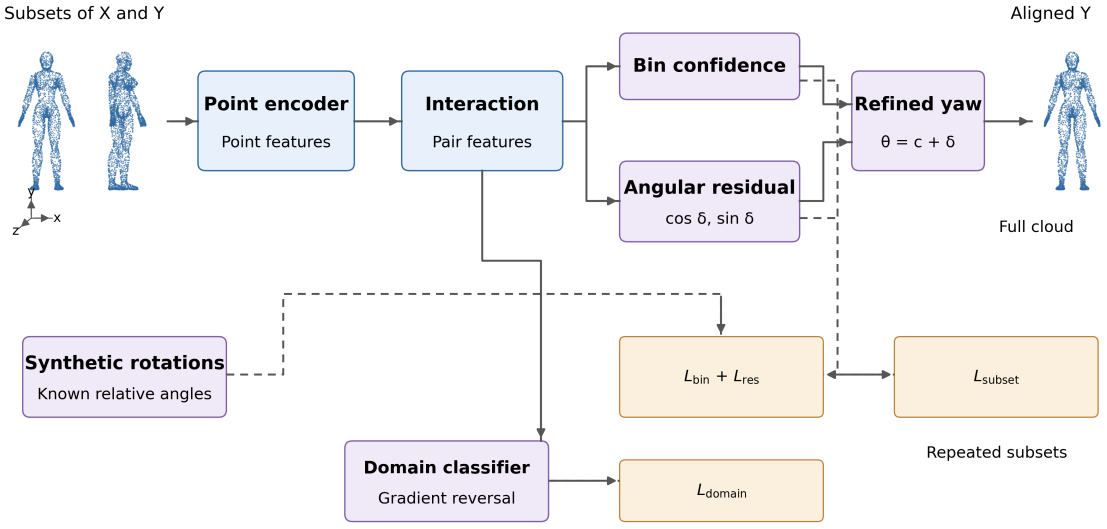


FIGURE 3.3: Relative orientation estimation. Pairwise point features predict angular-bin confidence and a continuous residual. Synthetic rotations supervise angle prediction, repeated point subsets supply consistency supervision, and a domain discriminator connects synthetic and cross-shape pairs. The predicted yaw aligns the full target point cloud.

3.3.2 Continuous angular prediction

Overlapping angular bins provide coarse orientation hypotheses. For each bin with centre c_k, the network predicts confidence p_k and a normalized residual vector r_k. Following the MultiBin principle (Mousavian et al., 2017), the residual represents the displacement from the bin centre through cosine and sine. The predicted angle is

$$k^* = \arg\max_k p_k, \qquad \hat{\theta} = c_{k^*} + \operatorname{atan2}(r_{k^*,2}, r_{k^*,1}). \tag{3.5}$$

The residual supports continuous alignment between bin centres. Overlap allows a boundary angle to be represented by neighbouring bins. The inverse predicted rotation aligns all target points with the source frame.

3.3.3 Orientation supervision

Synthetically rotating a shape produces a paired observation with a known relative angle. Forward and inverse directions supervise the bin probabilities and residuals. Independently sampled observations of the same pair provide a consistency objective on both outputs. A gradient-reversal discriminator distinguishes synthetic same-shape pairs from cross-shape pairs (Ganin et al., 2016); focal loss weights this domain classification (Lin et al., 2017).

The orientation objective combines angular classification, residual prediction, subset consistency and domain adaptation. Alignment is detached when supplying coordinates to the correspondence branch, so OEM is trained by its dedicated supervision while ASMAE is trained by representation objectives.

3.4 ASMAE Representation Learning

3.4.1 Shared geometric encoder

ASMAE maps the input heat descriptors and XYZ coordinates to point embeddings. A feature projection and a positional encoding initialize the point tokens. Local attention aggregates neighbouring tokens, with relative geometric direction supplying an attention bias. Both shapes pass through the same encoder parameters, placing their descriptors in a common feature space.

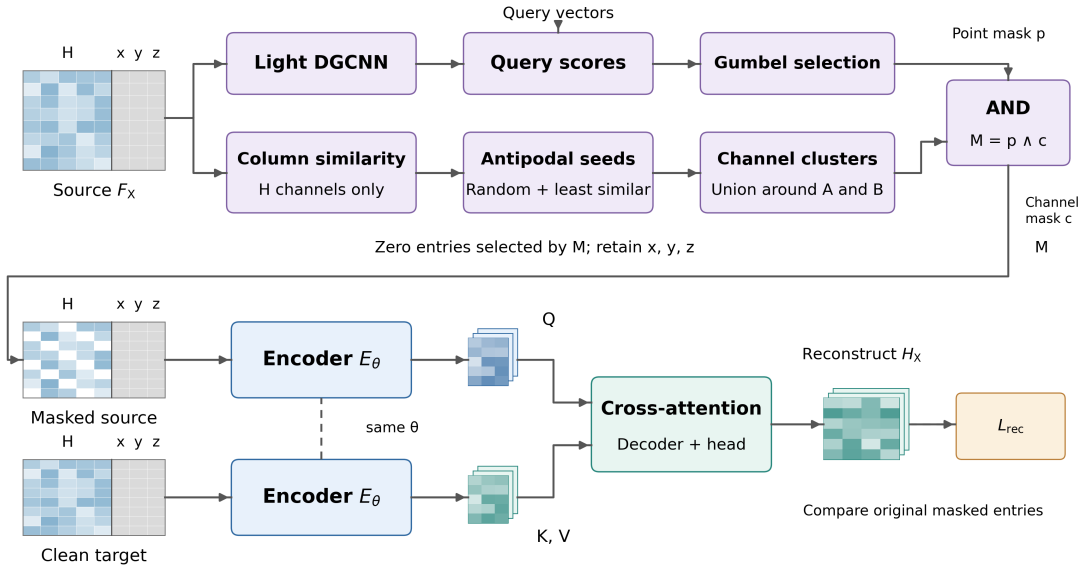


FIGURE 3.4: ASMAE and its mask generator. Point selection and descriptor-channel selection jointly define the mask; XYZ remains visible. Masked source tokens query clean target features for reconstruction, also performed in the reverse direction. Both E_θ blocks denote the same encoder parameters; the mask generator has its own Light DGCNN.

3.4.2 Descriptor masking and cross-reconstruction

The AAMG mask generator has two selection branches, shown in Figure 3.4. A Light DGCNN extracts features for point selection. Query vectors score these features, and hard Gumbel selection chooses point locations. The union of selected locations gives a binary point mask p_i.

The second branch computes cosine similarity between descriptor columns, excluding XYZ. It selects a random seed channel and a channel least similar to that seed, then combines channels similar to each seed. This gives a channel mask c_j. Their intersection defines the entries to hide:

$$M_{ij} = p_i \wedge c_j, \qquad \widetilde{F}_{ij} = \begin{cases} 0, & M_{ij} = 1, \\ F_{ij}, & M_{ij} = 0. \end{cases} \tag{3.6}$$

Coordinate channels remain visible, and all point positions are retained. The Light DGCNN belongs to the mask generator; the shared ASMAE encoder E_θ processes the resulting masked or clean feature matrices.

The masked source representation supplies decoder queries. Clean target embeddings supply keys and values, allowing the decoder to use cross-shape context to reconstruct source descriptors. Exchanging the source and target gives bidirectional reconstruction. This paired masked-learning design is motivated by AMIGO (Deng and Lu, 2025). Separate clean encoder passes supply embeddings for the auxiliary objectives and matching.

3.5 Visual Feature Module

The visual branch projects the point cloud into multiple depth-based views. A frozen DINOv2 backbone extracts patch descriptors (Oquab et al., 2024). Visibility testing associates image samples with surface points, and aggregation across views produces a point descriptor d_i with observation confidence c_i. Neighbourhood smoothing shares visual context locally.

A prediction head maps each geometric embedding to a visual descriptor $\hat{d}_i$. Distillation weights the lifted observations by confidence:

$$L_{\text{DINO}} = \frac{\sum_i c_i \left[1 - \cos(\hat{d}_i, d_i)\right]}{\sum_i c_i}. \tag{3.7}$$

The objective is evaluated on both shapes. At matching time, observation confidence controls the blend of observed and predicted visual features. The visual branch therefore supplies both training supervision and an inference descriptor, following the use of lifted visual information in non-rigid matching (Chen et al., 2025).

3.6 Representation Objectives

The representation objectives supervise complementary properties of the shared encoder. Reconstruction uses masked entries, geometric objectives use clean paired embeddings, self-identity uses an augmented observation of the same shape, and visual distillation uses lifted DINOv2 descriptors.

3.6.1 Reconstruction and feature separation

The reconstruction loss measures the absolute error on masked descriptor entries in both directions. A separation term discourages excessive similarity between distinct points. For normalized embeddings z_i, its form is

$$L_{\text{sep}}(Z) = \frac{1}{N(N-1)} \sum_{i \neq j} \max(z_i^\top z_j - m, 0), \tag{3.8}$$

where m is the similarity margin. Together, these objectives preserve reconstructive information and discrimination between point features.

3.6.2 Cycle and intrinsic consistency

Feature similarities define directional soft maps

$$P_{XY} = \text{softmax}_j(Z_X Z_Y^\top / \tau), \tag{3.9}$$

with temperature τ. The reverse map is constructed analogously. The cycle objective measures coordinate recovery after mapping in both directions:

$$L_{\text{cycle}} = \text{MSE}(P_{XY} P_{YX} X, X) + \text{MSE}(P_{YX} P_{XY} Y, Y). \tag{3.10}$$

For intrinsic consistency, the soft map transports the target biharmonic embedding to source indexing, $\widetilde{\Psi}_X = P_{XY}\Psi_Y$. The directional distortion is

$$L_B(X,Y) = \underset{i,j}{\text{mean}} \left| d_B^X(i,j) - \|\widetilde{\psi}_X(i) - \widetilde{\psi}_X(j)\|_2 \right|. \tag{3.11}$$

Averaging the two directions encourages matched points to preserve the intrinsic relationships within each shape.

3.6.3 Local orientation consistency

Neighbouring source points form triplets with directed area vectors. Their soft-mapped triplets define corresponding vectors in the aligned target frame. An angular agreement term compares the normalized vectors. This supplies a local directional constraint alongside the distance-based objective.

3.6.4 Self-identity supervision

The self-identity branch uses the same ASMAE encoder E_θ as reconstruction and clean feature extraction. It reuses the clean source embedding $Z_X = E_\theta(F_X, X)$ and computes an additional embedding for an augmented source subset. Both passes share parameters and receive gradients from the identity objective.

A clean shape and an augmented subset share point identities. Rotation, scaling and coordinate perturbations change the observation while retaining the correspondence of sampled vertices. An InfoNCE objective (van den Oord et al., 2018) treats the matching point as a positive candidate:

$$L_{\text{self}} = -\frac{1}{|Q|} \sum_{i \in Q} \log \frac{\exp(z_i^\top z'_{\pi(i)}/\tau_s)}{\sum_j \exp(z_i^\top z'_j/\tau_s)}. \tag{3.12}$$

Here Q contains queried source identities retained in the subset, $\pi(i)$ is the subset position of the same original vertex, and τ_s controls the contrastive distribution. Embeddings are normalized before comparison. Cross-entropy rewards the matching identity against all other points in the augmented subset. The objective is evaluated on the source member of each ordered training pair.

3.6.5 Joint optimization

Representation learning combines the objectives through weights λ_j:

$$L_{\text{rep}} = \sum_{j \in \mathcal{J}} \lambda_j L_j, \quad \mathcal{J} = \{\text{rec}, \text{sep}, \text{cycle}, B, \text{orient}, \text{self}, \text{DINO}\}. \tag{3.13}$$

Training first establishes relative orientation and then introduces the representation objectives progressively. The overall objective combines L_{rep} with OEM and domain supervision. This separates the supervision of alignment from the learning of point descriptors while connecting both through the aligned inputs.

3.7 Heat-Anchored Deformation Matching

HADM converts geometric and visual embeddings into a dense point map through orientation selection, anchor propagation and deformation refinement.

3.7.1 Orientation selection and feature fusion

Predictions from independently sampled point subsets contribute orientation votes. Candidate alignments include strong angular modes and a half-turn alternative. For each candidate, the target is aligned and encoded. Intrinsic anchor agreement and feature similarity jointly score the candidate, and the selected alignment supplies the pair for refinement.

Observed and predicted visual descriptors are blended using their confidence. A shared dimensionality reduction places the visual features of both shapes in a common compact representation. Normalized geometric and visual feature blocks are then fused for correspondence search.

3.7.2 Intrinsic anchor consistency

Mutual feature matches propose anchor pairs. Similarity margins rank their confidence, while spatial coverage in biharmonic space distributes candidates across the shape. Agreement between candidate pairs is measured by

$$W_{uv} = \exp\left[-\frac{(d_B^X(a_u, a_v) - d_B^Y(b_u, b_v))^2}{2\sigma_c^2}\right]. \tag{3.14}$$

A spectral score identifies mutually compatible candidates, and consistency filtering forms the anchor set. This uses the pairwise-compatibility principle of spectral matching (Leordeanu and Hebert, 2005).

3.7.3 Anchor propagation

Corresponding anchors define proximity descriptors on both shapes:

$$h_{j,\sigma}(x) = \exp\left[-\frac{\|\psi(x) - \psi(a_j)\|_2^2}{2\sigma^2}\right]. \tag{3.15}$$

These Gaussian responses encode position relative to the anchor set in biharmonic space. Combining them with point features transfers anchor information to other locations. Successive stages expand and refine the anchor set using this additional context.

3.7.4 Deformation and dense assignment

A similarity transformation estimated from the anchors initializes source alignment. A deformation graph then represents non-rigid adjustment through local node transformations (Sumner et al., 2007). Each point blends nearby transformations to obtain its deformed position. Positional agreement and anchor constraints guide the fit, while as-rigid-as-possible regularization encourages compatible motion between neighbouring graph nodes (Sorkine and Alexa, 2007). Refinement proceeds from a coarse graph to finer spatial detail.

The final map combines fused feature similarity with deformed positional agreement:

$$T(i) = \arg\max_{j} \left[G_X(i)^\top G_Y(j) - \lambda_p \|\widetilde{x}_i - y_j\|_2^2 \right]. \tag{3.16}$$

Here G contains the fused features and anchor-relative descriptors, and $\widetilde{X}$ is the deformed source. The resulting point indices support attribute transfer between shapes, as illustrated by the colour-transfer operation in Figure 3.1.

Chapter 4

Experiments and Results

4.1 Datasets

FAUST (Bogo et al., 2014), SCAPE (Anguelov et al., 2005), SHREC'19 (Melzi et al., 2019) and SURREAL (Varol et al., 2017) provide human-shape data for studying correspondence under variation in pose and identity. FAUST contains registered human scans, SCAPE models articulated body deformation, SHREC'19 addresses correspondence between human shapes, and SURREAL supplies synthetic human observations. These benchmarks connect geometric matching to the anatomical structure of deformable bodies.

4.2 Correspondence Evaluation

For a source point i, let $T(i)$ be its predicted target and $T^*(i)$ the reference match. Normalized geodesic error measures their separation on the target surface:

$$e_i = \frac{d_Y(T(i), T^*(i))}{\sqrt{\text{area}(Y)}}. \tag{4.1}$$

The mean over evaluated vertices gives a pair score, and averaging pair scores gives the aggregate error. Lower error indicates closer agreement with the reference correspondence. Surface-area normalization supports comparison across different shape scales.

Colour transfer provides a complementary view of the dense map. A source colour field assigns distinct colours to different regions, and correspondence transfers these attributes to the target. Consistent placement of colour regions allows inspection of anatomical alignment across changes in pose. The same visualization convention supports comparisons between the pipeline and its component variants.

Chapter 5

Conclusion

This report presents a correspondence pipeline that connects point-cloud representation learning with relative orientation and deformation-guided matching. Spectral heat descriptors capture relationships over the point graph, while explicit XYZ channels provide coordinate context. The orientation module places the input pair in a compatible frame using angular classification and continuous refinement.

The shared ASMAE learns point features through masked reconstruction across the two shapes. Geometric consistency preserves spatial relationships, self-identity supervision retains point identity under augmentation, and DINOv2 distillation transfers visual information to the geometric representation. These objectives connect local discrimination with shape-wide structure.

HADM combines geometric and visual features to propose correspondences, identifies anchors through intrinsic consistency, and propagates their information across the shapes. Deformation refinement then links the feature representation to positional agreement. The resulting dense point map provides the basis for transferring surface attributes between differently posed shapes.

The experimental progression highlights the complementary roles of representation, orientation and matching. Distortion regularization partially reduced left–right ambiguity in qualitative inspection, while front–back ambiguity motivated relative orientation modelling. The resulting design uses a shared encoder, continuously refined alignment and anchor-guided deformation as connected modules of the correspondence system.

5.1 Ablation Overview

The experimental progression examines the representation, alignment and matching components of the correspondence pipeline. The following observations summarize the motivation for each group of experiments, with detailed descriptions in Appendix A.

The initial student–teacher baseline studied correspondence learning without explicit geometric descriptors. A moving-average teacher supplied consistency targets to the student. The baseline and its reconstruction formulation are described in Section A.1.

Adding XYZ coordinates supplied positional context alongside the point descriptors. Protecting these channels during masking retained a spatial reference for reconstruction. Section A.2 explains coordinate conditioning and its connection to the choice of reference frame.

DINOv2 introduced a complementary visual cue through descriptors lifted from projected views. This experiment examined visual feature matching before the later distortion and orientation changes; its construction is described in Section A.3.

Distance-preservation regularization reduced left–right ambiguity to some extent in qualitative inspection. Front–back ambiguity motivated the subsequent directional

and alignment experiments. Section A.4 describes the distortion objective and these observations.

The descriptor experiments combined heat-based features, WKS and normals estimated from point-cloud neighbourhoods. These were point-estimated normals, not mesh normals. Section A.5 discusses the complementary spectral and directional information.

Canonicalization explored a shared reference frame before correspondence estimation. Relative orientation then used pairwise feature interaction through the SE-ORNet orientation module. These approaches are detailed in Sections A.6 and A.7, respectively.

The orientation variants examined coordinate noise, independently sampled points, arbitrary-angle augmentation and boundary avoidance. Overlapping bins and continuous residuals supplied a finer angular representation. Sections A.8 and A.9 describe these experiments and subset consistency.

The shared-encoder design removed the student–teacher arrangement following observed degradation and combined intrinsic distortion, self-identity supervision and visual distillation. HADM connected the features to anchor consistency and deformation refinement. Section A.10 describes this integration.

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Appendix A

Experimental Development

This appendix explains the experimental progression summarized in Section 5.1. The variants examine how representation, orientation and geometric matching interact when estimating correspondence between deformable shapes.

A.1 Student–Teacher Baseline

The initial experiment used student–teacher learning without explicit geometric descriptors. The student learned from paired shape observations, and a moving-average teacher supplied a consistency target. This follows the broader principle of stabilizing representation learning through a slowly updated target model (Tarvainen and Valpola, 2017).

Masked reconstruction related the source to a reference shape. The associated global matching objective used an entropically regularized transport formulation (Cuturi, 2013). This stage established a representation-learning baseline for examining the contribution of additional geometric and visual cues.

A.2 Adding Coordinates

Explicit XYZ channels introduced positional information into the point representation. Coordinates describe the location of a point within the shape and complement spectral descriptors that can be similar at symmetric locations. The input matrix combined heat-based features with normalized spatial coordinates.

Coordinate conditioning also made the reference frame relevant to feature learning. The same anatomical point can change its coordinates under rotation, and different poses can place corresponding body parts in different positions. This connection between position and orientation motivated the alignment experiments. Protecting coordinate channels during masking preserved the positional context of hidden descriptor entries.

A.3 Adding DINOv2 Features

The visual-cue experiment added DINOv2 features to the matching representation (Oquab et al., 2024). Projected shape views supplied image descriptors, which were lifted to their visible points. Aggregation across views associated each point with visual context complementary to the geometric encoder.

This stage focused on adding the visual cue before the later distortion and orientation changes. Visibility handling controlled which image observations contributed to a point descriptor. The feature-matching path combined the lifted representation with learned point features. Visual distillation, described later, extended this connection into the training objectives.

A.4 Distortion and Symmetry

A distance-preservation objective constrained the spatial relationships induced by a soft correspondence. If P maps source indices to target points, the transported positions are $\widehat{Y} = PY$. The Euclidean distortion objective compares distances between source points and their transported counterparts:

$$L_{\text{dist}} = \underset{i,j}{\text{mean}} \left| \|x_i - x_j\|_2 - \|\widehat{y}_i - \widehat{y}_j\|_2 \right| . \tag{A.1}$$

In qualitative inspection, this term corrected left–right ambiguity to some extent. It encouraged groups of matched points to preserve their spatial relationships, rather than making each assignment independently. Front–back ambiguity then became a central focus: a large region could have coherent matches while being associated with the opposite facing direction. The following experiments examined additional directional cues and explicit orientation estimation.

A.5 HKS, WKS and Point Normals

A richer geometric representation combined heat-based features, WKS, coordinates and normal components. HKS and WKS describe complementary aspects of the spectrum (Sun et al., 2009; Aubry et al., 2011). Normal components supplied a local directional cue alongside these descriptors.

The normals were estimated from point-cloud neighbourhoods; they were not mesh normals. Local covariance analysis identified the direction of least variation as a normal axis. Centroid-based orientation and neighbourhood sign propagation supplied variants for assigning its direction. The normal and coordinate channels were preserved during reconstruction masking.

The experiments also examined an orientation penalty based on normal compatibility and a global proper-rotation criterion based on a fitted transformation. Reflecting the coordinate frame provided another way to examine front–back and left–right alternatives. These variants connected local descriptors with the direction implied by a candidate correspondence.

A.6 Canonical Orientation

Canonicalization predicted a common frame for each shape before matching. A global point-cloud representation produced a continuous rotation parameterization (Zhou et al., 2019), which was converted to an orthonormal transformation and applied to the centred points. Synthetic rotations supplied a known relationship between transformed observations.

Equivariance, canonical-coordinate agreement and consistency supervision encouraged the same shape to retain a common canonical interpretation under rotation. The approach is related in purpose to shape–pose disentanglement (Katzir et al., 2022). The experiments highlighted the importance of choosing compatible frames across different articulated poses, motivating pairwise relative orientation.

A.7 SE-ORNet Orientation Estimation

The pairwise orientation module estimates how the target is rotated relative to the source. Cross-shape feature interaction connects the two point representations before angular prediction. The SE-ORNet formulation (Deng et al., 2023) supplied the foundation for synthetic rotation supervision and domain adaptation.

A synthetically rotated copy of a shape supplies known forward and inverse angles. Real pairs of distinct shapes contribute to domain adaptation, which aligns their

pair representations with those of synthetic observations. Group normalization (Wu and He, 2018) stabilizes intermediate feature normalization. The estimated relative rotation is applied before coordinate-conditioned correspondence learning.

The orientation experiments also examined the relationship between learning alignment and learning correspondence. Orientation-focused optimization established the alignment task before progressively introducing the representation objectives. This separates their supervision while connecting their outputs through the aligned point coordinates.

A.8 Noise, Sampling and Angular Boundaries

A.8.1 Coordinate noise and point sampling

Coordinate perturbations tested orientation learning under small changes in the observed positions. Scaling perturbations by cloud size made them relative to the input geometry. Independently sampled point subsets tested whether the pair representation retained the same orientation information when different points were observed.

Random sampling and spatially distributed sampling provided complementary subset strategies. The orientation prediction was applied to the full target cloud, preserving all points for dense correspondence. Reconstruction masking addressed a different question: it hid feature information while retaining the point locations and coordinate context for ASMAE.

A.8.2 Arbitrary angles and boundary avoidance

Angle-centre training associated synthetic rotations with discrete orientation classes. Arbitrary-angle augmentation extended the observations between class centres, exposing the classifier to continuously varying orientations. A boundary-avoidance variant restricted training rotations to neighbourhoods of the bin centres, excluding the regions where adjacent class assignments change.

These experiments separated two issues: sensitivity to class boundaries and quantization of a continuous angle. Avoiding boundary regions altered the training distribution, while continuous residual prediction supplied an explicit way to represent the displacement within a bin.

A.9 Continuous Angular Refinement

The refinement experiment combined overlapping angular bins with a residual for each bin, following MultiBin (Mousavian et al., 2017). A confidence score selected an orientation hypothesis, and a cosine–sine vector described its continuous correction. Overlapping support allowed nearby bins to share supervision for boundary angles.

Independent observations of the same pair supplied subset consistency. The objective compared their bin distributions and the residual predictions associated with corresponding bins. This retained separate orientation hypotheses while encouraging agreement under point resampling. Multiple subset predictions also supplied votes during inference.

A.10 Shared Representation and HADM

The student–teacher arrangement was removed after it was observed to degrade correspondence during development. A shared ASMAE model learned both shapes

directly, and the associated global transport objective was removed in the same redesign. Reconstruction was combined with intrinsic distortion, self-identity supervision and visual distillation.

Biharmonic distortion compares distances in a spectral embedding (Lipman et al., 2010). Self-identity supervision associates the same vertices across augmented observations through an InfoNCE objective (van den Oord et al., 2018). DINOv2 distillation transfers lifted visual information into the learned point representation, with observation confidence weighting the contribution of each point.

HADM connects these representations to geometric matching. Fused features propose anchors, intrinsic distance agreement filters the proposals, and anchor-relative descriptors propagate spatial context. Orientation hypotheses are evaluated using feature and anchor evidence. A deformation graph then refines source positions under data, anchor and local rigidity constraints (Sumner et al., 2007; Sorkine and Alexa, 2007). Dense assignment combines the resulting positional agreement with feature similarity. This integration links relative alignment, representation learning and non-rigid refinement within a single correspondence pipeline.